

⚠ This backtest uses only data available at each test date – no lookahead bias. The engine runs exactly as in live trading, sliced to the test date. Expect 5-15 minutes for a full run (46 test dates × full analogue search).

Pattern window

12 months (252 days)

Run Validation Backtest

Phase 1+2 Direction

Panel 3 Sector Rotation

Tests Panel 3's method: given the top analogues at each date, does investing in the sector that historically led in those analogue periods outperform SPY? Only uses post-1998 ETF data (no proxy baskets). Test window: 2002-2022.

Pattern window

12 months (252 days)

Run Panel 3 Validation

5s elapsed

Complete in 6s

41 / 41

BEAT SPY RATE

43.2%

vs 47.7% random baseline

LIFT OVER RANDOM

-4.5pp

percentage point improvement

AVG RELATIVE RETURN

-3.2%

recommended sector vs SPY

AVG SECTOR RANK

5.4 / 9

1=best, 9=worst out of 9 sectors

GOT BEST SECTOR

16.2%

exactly matched top sector

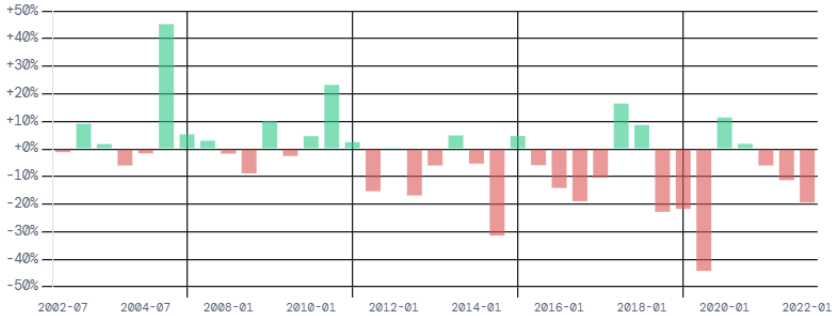
TEST DATES

37

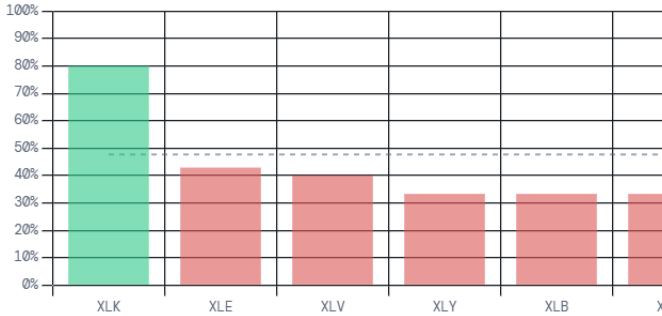
4 skipped (no ETF data)

Panel 3 underperforms random sector selection — the method needs improvement.
The recommended sector beat SPY only 43.2% of the time vs 47.7% baseline. The analogue-based sector prediction may be counterproductive.

Recommended sector actual relative return vs SPY over time



Beat SPY rate by recommended sector



ACCURACY BY RECOMMENDED SECTOR

Sector	Name	Times recommended	Beat SPY %	Avg relative return
XLK	Technology	5	80%	+2.9%
XLE	Energy	14	42.9%	-5.6%
XLV	Healthcare	5	40%	-2.5%
XLY	Consumer Disc	6	33.3%	-2.6%
XLB	Materials	3	33.3%	-0.8%
XLU	Utilities	3	33.3%	-7.3%
XLF	Financials	1	0%	-1.6%

Test date	Top recommendation	P3 score	Actual rel 12mo	Beat SPY?	Actual best	Sector rank	Macro
2002-07-01	XLY	22.0%	-1.2%	X No	XLK	4/9	Fed 1.7% YC 1.97%
2003-01-01	XLB	24.0%	+9.1%	✓ Yes	XLB	1/9	Fed 1.2% YC 2.22%
2003-07-01	XLK	32.5%	+1.8%	✓ Yes	XLE	4/9	Fed 1.0% YC 2.26%
2004-01-01	XLK	25.1%	-6.0%	X No	XLE	8/9	Fed 1.0% YC 2.43%
2004-07-01	XLF	17.5%	-1.6%	X No	XLE	3/9	Fed 1.3% YC 1.93%
2005-01-01	XLE	25.0%	+45.3%	✓ Yes	XLE	1/9	Fed 2.3% YC 1.16%
2006-01-01	XLU	31.6%	+5.3%	✓ Yes	XLU	1/9	Fed 4.3% YC -0.02%
2006-07-01	XLE	24.6%	+3.0%	✓ Yes	XLB	4/9	Fed 5.2% YC -0.01%
2008-01-01	XLE	19.0%	-1.7%	X No	XLP	6/9	Fed 3.9% YC 0.99%
2008-07-01	XLB	13.8%	-8.8%	X No	XLP	7/9	Fed 2.0% YC 1.38%
2009-01-01	XLY	12.1%	+10.2%	✓ Yes	XLK	3/9	Fed 0.1% YC 1.49%
2009-07-01	XLB	10.5%	-2.6%	X No	XLY	6/9	Fed 0.2% YC 2.50%
2010-01-01	XLE	29.0%	+4.6%	✓ Yes	XLY	3/9	Fed 0.1% YC 2.71%
2010-07-01	XLE	36.3%	+23.2%	✓ Yes	XLE	1/9	Fed 0.2% YC 2.33%
2011-01-01	XLE	14.0%	+2.5%	✓ Yes	XLU	5/9	Fed 0.2% YC 2.69%
2011-07-01	XLE	25.9%	-15.3%	X No	XLU	9/9	Fed 0.1% YC 2.72%
2012-01-01	XLK	8.5%	+0.4%	✓ Yes	XLF	4/9	Fed 0.1% YC 1.64%
2012-07-01	XLU	12.0%	-16.9%	X No	XLF	9/9	Fed 0.2% YC 1.34%
2013-01-01	XLE	30.0%	-6.0%	X No	XLY	6/9	Fed 0.1% YC 1.53%
2013-07-01	XLE	14.8%	+4.9%	✓ Yes	XLB	3/9	Fed 0.1% YC 2.16%
2014-01-01	XLY	7.2%	-5.3%	X No	XLU	7/9	Fed 0.1% YC 2.66%
2014-07-01	XLE	4.0%	-31.4%	X No	XLV	9/9	Fed 0.1% YC 2.11%
2015-01-01	XLV	2.6%	+4.7%	✓ Yes	XLY	3/9	Fed 0.1% YC 1.50%
2015-07-01	XLV	10.2%	-5.9%	X No	XLU	6/9	Fed 0.1% YC 1.74%
2016-01-01	XLV	4.1%	-14.1%	X No	XLE	9/9	Fed 0.3% YC 1.21%
2016-07-01	XLE	16.4%	-18.9%	X No	XLF	9/9	Fed 0.4% YC 0.87%
2017-01-01	XLU	7.1%	-10.4%	X No	XLK	8/9	Fed 0.7% YC 1.25%
2017-07-01	XLK	6.8%	+16.5%	✓ Yes	XLK	1/9	Fed 1.1% YC 0.93%
2018-01-01	XLV	9.2%	+8.7%	✓ Yes	XLV	1/9	Fed 1.4% YC 0.51%
2018-07-01	XLE	6.4%	-22.8%	X No	XLU	9/9	Fed 1.9% YC 0.33%
2019-01-01	XLE	29.3%	-21.7%	X No	XLK	9/9	Fed 2.4% YC 0.21%
2019-07-01	XLE	7.3%	-44.2%	X No	XLK	9/9	Fed 2.4% YC 0.25%
2020-01-01	XLY	7.5%	+11.4%	✓ Yes	XLK	2/9	Fed 1.6% YC 0.34%
2020-07-01	XLK	7.0%	+1.8%	✓ Yes	XLF	5/9	Fed 0.1% YC 0.52%
2021-01-01	XLV	7.8%	-6.0%	X No	XLE	6/9	Fed 0.1% YC 0.80%
2021-07-01	XLY	9.8%	-11.3%	X No	XLE	9/9	Fed 0.1% YC 1.23%
2022-01-01	XLY	4.7%	-19.4%	X No	XLE	9/9	Fed 0.1% YC 0.79%

